guide-for-data-science-peers

PMIP — Technical walkthrough for data science peers

Audience: classmates and faculty in a Master's of Data Science program
Product: Precious Metals Macro Intelligence Platform (PMIP)
Goal of this note: explain what was built, which models produce which numbers, how evaluation avoids look-ahead, and what is still a rule or a seed rather than a learned probability.

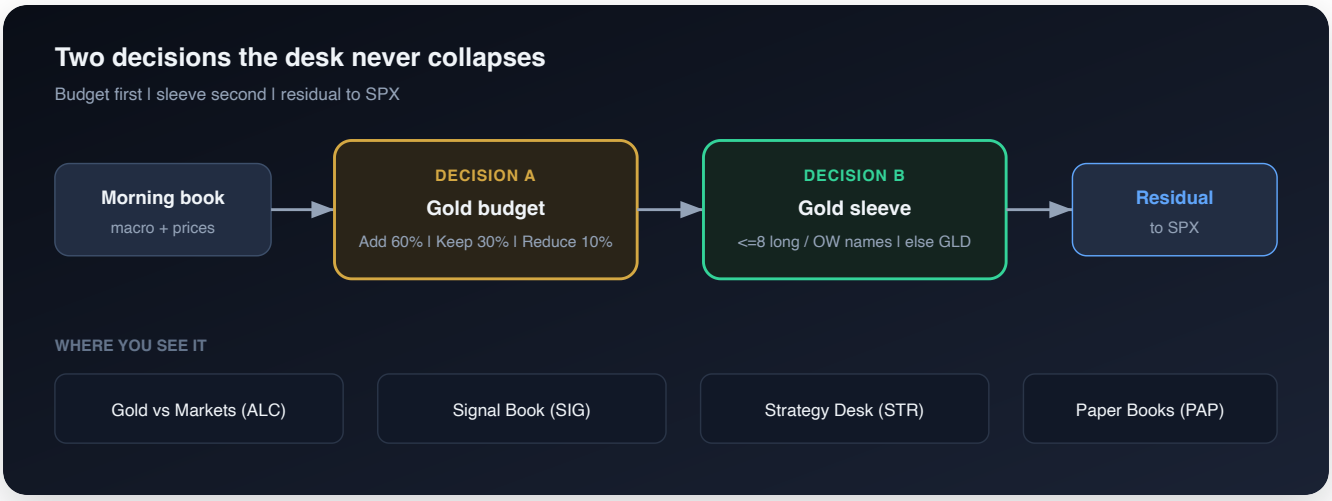
Related reading: Research note (citeable purged study), PRD, README.

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1. What problem this solves

A gold allocation book faces **two different decisions**:

1. **Gold budget** — Over the next few weeks, should gold exposure be larger or smaller versus equities (and the rest of a multi-asset book)?
2. **Gold sleeve** — Given that budget, how should it be split between bullion (e.g. GLD / futures) and miners / royalties?

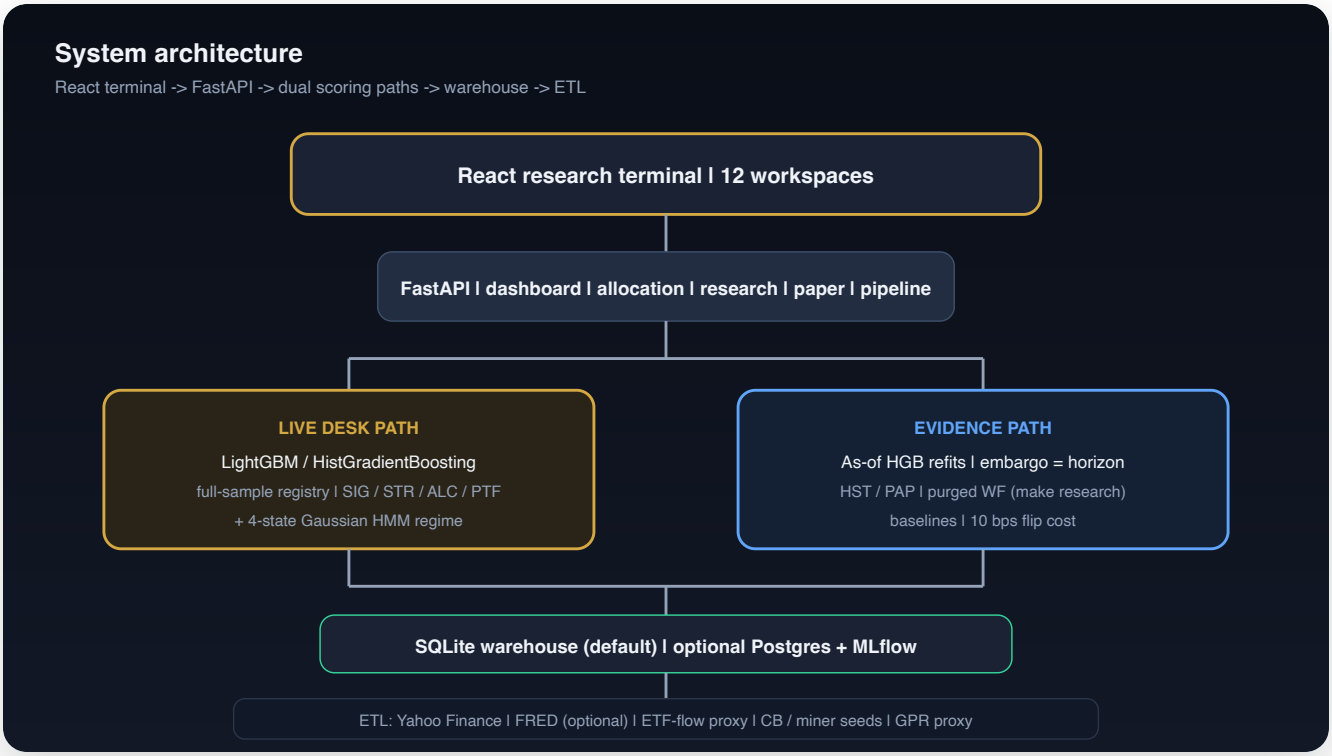


Most "AI for gold" demos collapse those into one score. PMIP keeps them separate, then adds an **evidence layer** (purged walk-forward + multi-horizon paper books) so you can tell classmates *what is allowed to mean what*.

It is a **local research desk**, not a trading system. It does not place orders.

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2. System architecture (one picture)

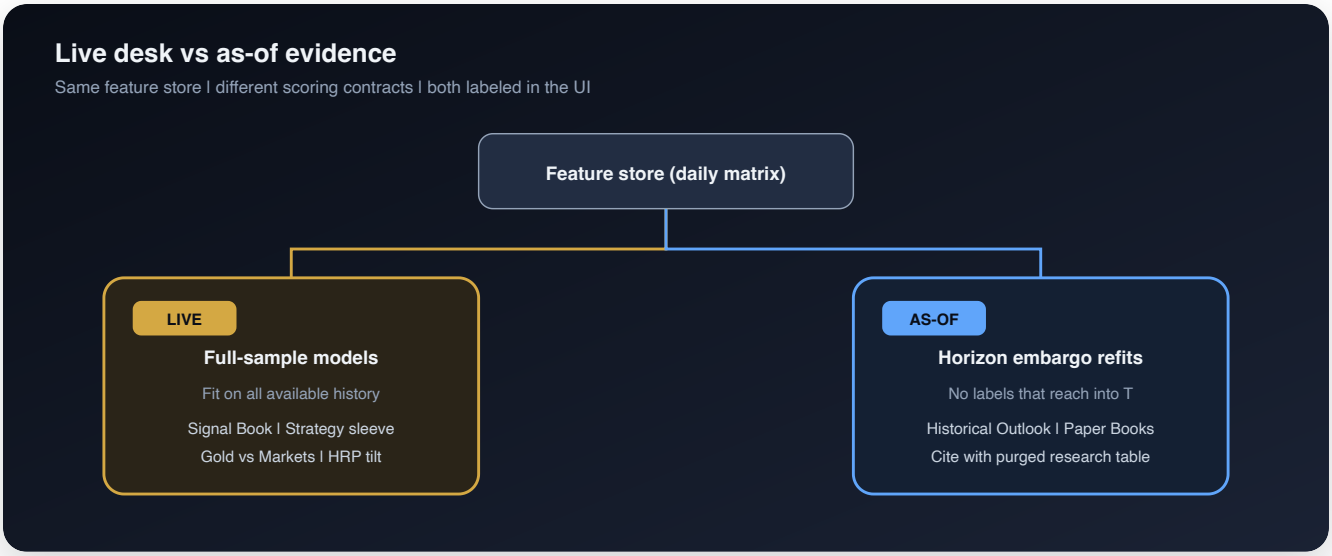


Layer	Choice
API	FastAPI + uvicorn
UI	React 18 + TypeScript + Vite + Recharts
Store	SQLite `data/pmip.db` (default); optional Postgres
ML	LightGBM when available; else sklearn `HistGradientBoosting`
Regime	`hmmlearn` Gaussian HMM (4 states)
Portfolio	Hierarchical risk parity (`riskfolio`) + prediction tilt
ETL	`yfinance`, optional `fredapi`

Run locally: `make install` → `make init-db` → pipeline (UI or `make pipeline`) → `make dev` → http://localhost:8000.

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3. The claim that matters most: live vs as-of



Path	Who uses it	How the model is fit	Version tag
Live / full-sample	Signal Book, Strategy sleeve, Gold vs Markets gold forecast, HRP expected-return tilt	Classifier + regressor trained on all available aligned history for that entity × horizon	`lgbm_h*` or `sklearn_h*`
As-of / embargo	Historical Outlook charts, Paper Books	At each score date *T*, refit using only rows whose forward labels are fully observed by *T* (train ends at least `horizon` trading days before *T*)	`asof_h{horizon}`

UI badges: **live** vs **as-of**. API field: `scoring_path` ∈ `{full_sample_live, asof_embargo}`.

Why peers should care: a dashboard that replays "today's model" on old feature rows overstates historical skill. The as-of path and the purged research study exist so you do not have to.

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4. Data: what enters the warehouse

Series	Source	Honesty label
Futures, ETFs, equities, SPX, VIX, oil	Yahoo Finance daily closes	Market data
Treasuries / TIPS / breakevens	FRED when `FRED_API_KEY` is set	Optional feed

Series	Source	Honesty label
Real yield fallback	Nominal − breakeven, or nominal − **2.25**	Tagged `tips` / `breakeven_spread` / `estimated`
ETF flows (GLD, IAU)	Derived from Yahoo AUM / share change	**Proxy / estimate**
Central-bank gold purchases	Seed monthly table (WGC-style)	**Seed / illustrative**
Miner fundamentals (AISC, production, EPS revision)	Seed quarterly	**Seed**
Geopolitical risk (GPR)	Composite of VIX / oil volatility	**Proxy** (not news NLP)

Rough scale: ~73 priced instruments; ~65 scored prediction targets; geared ETFs may be priced but are not primary model targets. As-of / paper liquid sleeve is a fixed list of ~14 liquid names (GLD, GDX, senior miners, royalties, etc.).

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5. Feature engineering

Built in `src/pmip/features/engineering.py` into a daily wide matrix.

Per name (price-based): trailing returns at 5 / 10 / 20 / 63 / 126 days; volatility 20d / 63d; distance to 200-day MA; 60d momentum and z-score; ATR(14); RSI(14); MACD.

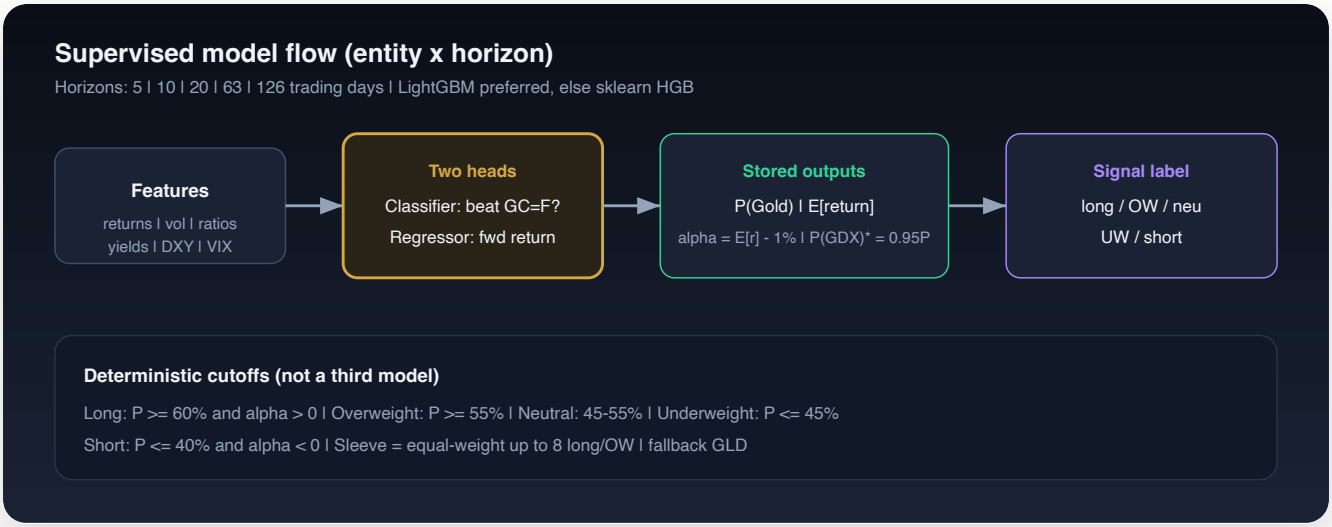
Cross-asset / macro block: gold vs oil / copper / SPX ratios and z-scores; real yield level, 20d trend, z-score; DXY 20d trend and z-score; VIX level and z-score; breakeven inflation and trend.

Miner block (where seed fundamentals exist): production / AISC surprises, revision momentum, 60d beta to gold.

Models see columns of the form `{entity}__{feature}` joined to the macro block.

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6. Supervised models (the metals book)



6.1 Learning problem

For each **entity** × **horizon** in `{5, 10, 20, 63, 126}` trading days:

Head	Target	Output used on screen
Binary classifier	1 if entity's forward return **beats gold futures (GC=F)** over the horizon	`prob_outperform_gold` → "P(Gold)"
Regressor	Entity's forward percent return over the horizon	`expected_return`

Display shortcuts (not second models):

- `prob_beat_gdx` = **0.95 ×** P(beat gold)
- `expected_alpha` = expected_return − **0.01** (flat 1% hurdle)

6.2 Algorithms

Preference order in `src/pmip/models/ml_backend.py`:

1. **LightGBM** — binary (AUC) / regression (RMSE); typical knobs: `num_leaves=31`, `learning_rate=0.05`, `feature_fraction=0.8`, `bagging_fraction=0.8`, ~200 rounds, `seed=42`.
2. Fallback **sklearn `HistGradientBoostingClassifier` / `Regressor`** — `max_iter=80`, `learning_rate=0.08`, `max_depth=5`, `random_state=42`.

As-of refits use a smaller HGB (`max_iter=40`, `max_depth=3`) so many dates remain practical.

Artifacts land under the model registry path as `{entity}_meta_h{horizon}.pkl` (+ model files). Versions are stamped on prediction rows.

6.3 Operational walk-forward (desk catalog)

During training, an adaptive purged walk-forward (embargo = horizon) records AUC / accuracy for the Model Validation catalog. **Deployed live desk models are still fit on the full available sample** for the morning signal — that is intentional and labeled.

6.4 Signal labels (deterministic cutoffs)

Label	Rule
Long	P ≥ 0.60 and alpha > 0
Overweight	P ≥ 0.55
Neutral	between 0.45 and 0.55
Underweight	P ≤ 0.45
Short	P ≤ 0.40 and alpha < 0

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7. Gold budget stance (rules + model spreads)

Gold vs Markets is not "the classifier's class." It is a **policy**:

1. Gold's 10-day expected return from the **live** GLD model (fallback: on-the-fly HGB on gold futures).
2. SPX / oil / copper each get a small HGB expected-return fit on the same feature store.
3. Macro gates: real yield > 2% → yield headwind; DXY 20d > +1% → dollar headwind; < –1% → support.
4. Regime gold bias is a **fixed label** for the HMM state (not a return forecast).

Stance	Rough rule	Gold sleeve budget in paper / Strategy
Add	Gold ahead of equities, no yield/dollar headwind, regime supportive	**60%**
Reduce	Gold behind equities and at least one headwind	**10%**
Keep	Mixed	**30%**

Residual weight → **SPX**. Sleeve inside the gold budget: equal-weight up to **8** long/overweight names; fallback **GLD**.

Peer note: Paper Books reuse the same Add/Keep/Reduce *budgets*, but paper stance compares gold's **as-of** expected return to **SPX's trailing return**, while Gold vs Markets uses a **10-day SPX model**. They can disagree; the UI says so.

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8. Regime model

`src/pmip/models/regime.py`:

- **Gaussian HMM**, 4 components, diagonal covariance, `n_iter=200`, `random_state=42`
- Features standardized: inflation / real yield levels and trends, DXY / oil / gold / SPX 20d trends, VIX, miner–gold spread
- States mapped to standing hypotheses (e.g. easing / dollar-weak → bullish gold; tightening → bearish)

Confidence on screen = max state probability. Gold/miner "bias" text is a lookup table, not an estimated expected return.

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9. Portfolio (HRP research weights)

`src/pmip/portfolio/optimizer.py`:

- Default blend: **60% hierarchical risk parity** + **40%** rank tilt from **live** 10-day expected returns
- Top 10 names by weight, renormalized, long-only
- Analytics: HHI, effective N (= 1 / Σw²), rough implied Sharpe using scaled 10d alpha / realized vol

This is a research allocation view, not an execution book.

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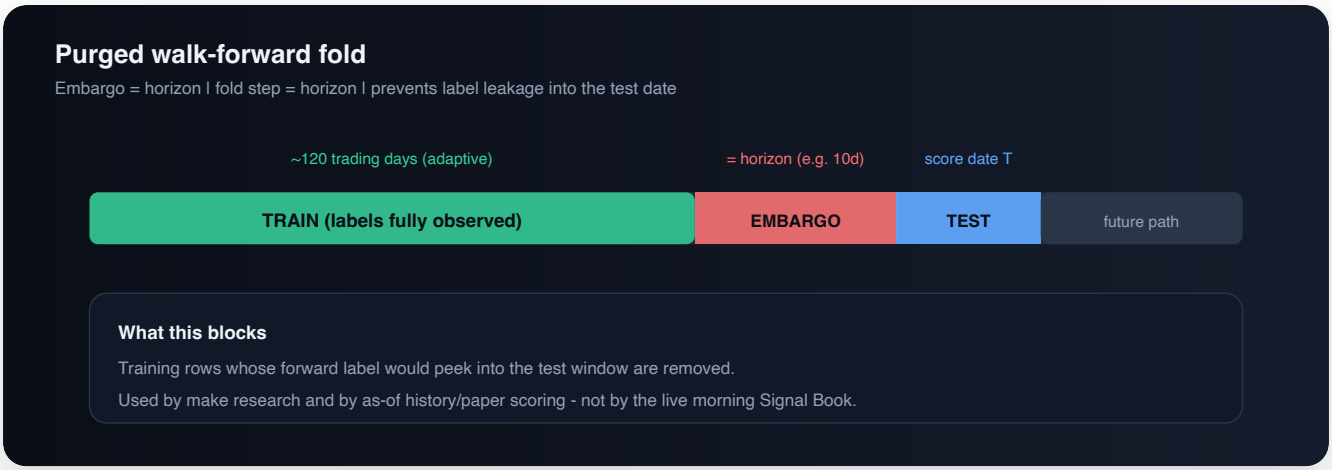
10. Paper books (shadow PnL)

For each horizon clock independently:

- Start **$100,000**
- Rebalance every **horizon** trading days when an **`asof_h*`** forecast exists that day
- Apply Add/Keep/Reduce budget → gold sleeve (≤8 long/OW) + SPX residual
- **10 bps** cost on dollars traded
- Mark to market vs buy-and-hold gold and SPX

This is the closest thing on the app to an economic simulation of the *as-of* scores. It is still a short-sample shadow book, not an audited live track record.

11. Purged research study (what to cite in class)



`make research` runs `src/pmip/research/evaluation.py` and rewrites:

- `docs/research-note.md`
- `docs/research/latest-results.md`
- `docs/research/results.json`

Two binary questions (horizon = 10 trading days):

1. Gold budget: hold **GC=F** or **SPX**?
2. Sleeve: hold **GDX** or **GC=F**?

Design choices peers usually ask about:

Choice	Implementation
Label leakage	Embargo = horizon; train folds end ≥ horizon before test
Overlapping returns	Fold step = horizon (non-overlapping economic path)
Baselines	Always-entity, always-benchmark, 20d relative momentum, logistic on a small macro set
Boosted model	HGB depth 3 / 40 iter; top 20 |corr| features *inside* each train fold
Costs	10 bps when the chosen asset flips
Calibration	5 probability bins vs realized hit rate
Stability	Logistic agreement after dropping real-yield and DXY columns

Important: cite this table for "does the timer earn its switches?" Do **not** cite Signal Book live P(Gold) as a track record. On the short sample used when the note was last generated, simple baselines often beat the booster on wealth — that negative / mixed finding is part of the honesty of the project.

Unit tests cover embargo structure and live-vs-as-of contracts (`tests/`).

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12. Strategy Desk (composition, not a new model)

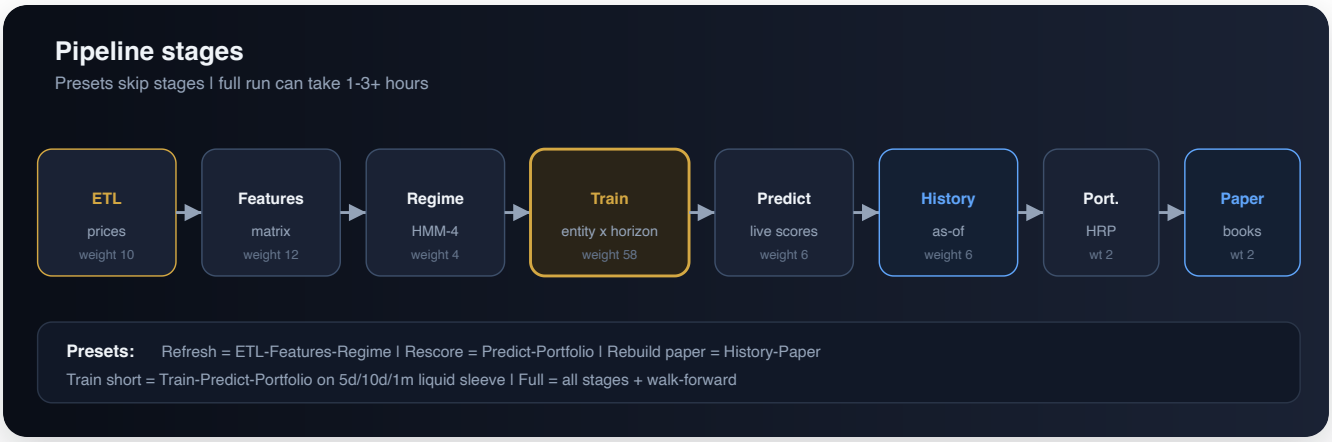
Rule-based playbooks composed from:

- Current Gold vs Markets stance → budget recipe
- Live 10d signal book → ≤8 long/OW sleeve
- Best paper clock by risk-adjusted score
- Regime overlay / defensive residual when headwinds fire

Confidence and ranking scores are heuristics. Caveat copy: research suggestion, not an order ticket.

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13. Pipeline (how data becomes a briefing)



Weighted stages: ETL → features → regime → train → predict → history (as-of) → portfolio → paper.

UI presets: refresh prices; rescore signals; train short horizons; rebuild paper; full run (can take 1–3+ hours). Sync reloads the warehouse without re-running ETL/train.

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14. Workspaces (map for a demo)

Code	Workspace	What a DS peer should look for
CMD	Command Center	Morning briefing; check whether insights overclaim seeds
STR	Strategy Desk	Composition of live sleeve + as-of paper evidence
ALC	Gold vs Markets	Policy rules + model spreads; real-yield source badge
MTL	Metals Book	Universe coverage and unpriced gaps
SIG	Signal Book	Live path badge; P(Gold) / alpha cutoffs; CSV export
MDL	Model Validation	**Cite purged table first**; catalog WF is operational
PAP	Paper Books	As-of only; costs; stance ≠ ALC object
PTF	Portfolio	HRP + live tilt
RSH	Research Desk	HMM regime history + knowledge-graph hypotheses
HST	Historical Outlook	As-of probability paths only
XPL	How It Works	Method + data-source inventory
RSK	Risk Monitor	Proxy / estimate / seed badges

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15. Known limits (say these out loud)

- Historical sample is short (~2 years of daily data in a typical local warehouse) versus institutional backtests.
- Without FRED, real yields are estimated.
- CB gold and miner fundamentals are seeds.
- GPR and ETF flows are proxies / estimates.
- Dual scoring paths must stay labeled; live ≠ paper.
- P(GDX) and expected alpha are display transforms.
- No broker, auth, SLA, or licensed market-data stack in v1.

Deferred on purpose (see PRD): model-only paper book beside the policy book; required FRED “production” gate; seed quarantine.

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16. Suggested 5-minute peer demo script

1. Open **How It Works** — two decisions + live vs as-of.
2. **Gold vs Markets** — show Add/Keep/Reduce drivers.
3. **Signal Book** — live badge, one horizon tab, one long name.
4. **Model Validation** — purged table + “what we would cite.”
5. **Paper Books** — as-of path, 10 bps, separate clocks.
6. End on limits: short sample, seeds, no order routing.

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17. Reproduce the evidence layer

```
make research   # purged studies → docs/research/
make test       # includes as-of embargo + live/as-of contracts
make verify     # broader health check when configured
```

If you only remember one sentence for peers: **PMIP is a claim-aware metals research OS — full-sample models for today’s briefing, purged as-of paths and a costed walk-forward for anything you would put in a paper.**